Thermodynamic description of the Al–Ni system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Al–Ni binary system as calculated with pycalphad from the SGTE dataset AlNi-14Zhu.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L + (Ni) → Ni3Al, lies at 1643.1 K.

1. Introduction

The Al–Ni system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as intermetallic rich, order disorder, congruent compound. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of J. Zhu, M.S. Titus, T.M. Pollock, J. Phase Equilib. Diffus., 35, 595-611(2014). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

The melting point of Al reproduced by the dataset is 939.4 K. For Ni, the liquidus meets the axis at 1728.2 K.

2. Phase diagram and invariant reactions

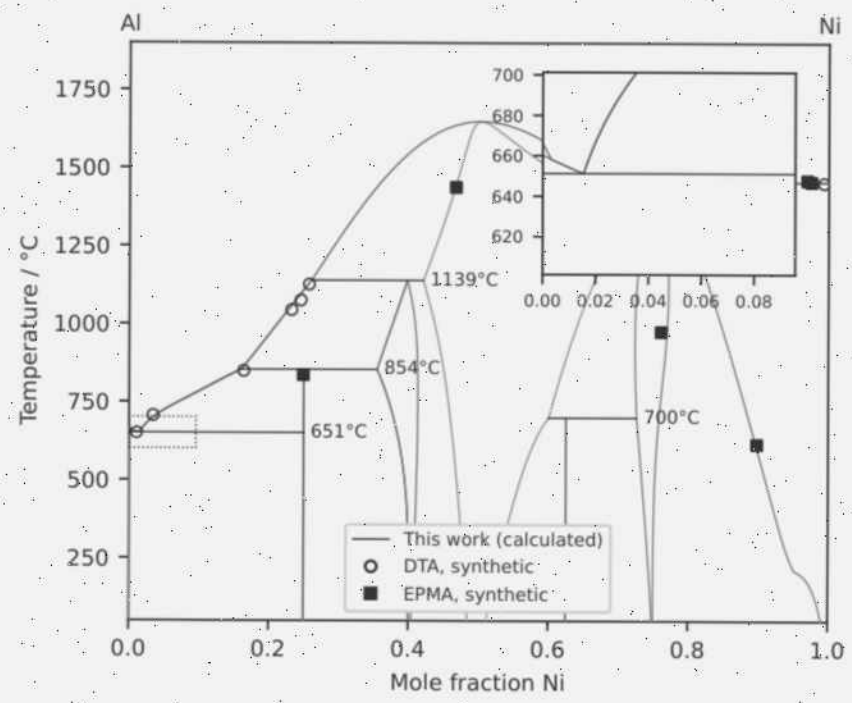


Fig. 1. Calculated Al–Ni phase diagram with synthetic experimental-style points (inset: enlarged view near the lowest invariant).

At 1643.1 K the peritectic L + (Ni) → Ni3Al takes place; the L phase then contains a mole fraction of 0.759. At 1642.3 K the eutectic L → NiAl + Ni3Al takes place; the L phase then contains a mole fraction of 0.744. At 1412.2 K the peritectic L + NiAl → Al3Ni2 takes place; the L phase then contains a mole fraction of 0.261. Peritectic

equilibrium L + Al3Ni2 → Al3Ni is found at 1127.0 K and a mole fraction of 0.160 (L). The dataset places a peritectoid (NiAl + Ni3Al → Al3Ni5) at 973.4 K; the composition of Al3Ni5 is x(Ni) = 0.625. At 924.4 K the eutectic L → (Al) + Al3Ni takes place; the L phase then contains x(Ni) = 0.016. (Ni) melts congruently at 1728.4 K and x(Ni) = 0.988. Ni3Al melts congruently at 1642.7 K and x(Ni) = 0.751.

Table 1. Invariant reactions and congruent points, calculated (Al–Ni).

Reaction	Type	T (°C)	x(Ni), 1	x(Ni), 2	x(Ni), 3
L → NiAl	congruent melting	1649.1	0.503	0.503	–
L → (Ni)	congruent melting	1455.2	0.988	0.988	–
L + (Ni) → Ni3Al	peritectic	1369.9	0.759	0.786	0.761
L → Ni3Al	congruent melting	1369.6	0.751	0.751	–
L → NiAl + Ni3Al	eutectic	1369.2	0.744	0.712	0.746
L + NiAl → Al3Ni2	peritectic	1139.1	0.261	0.420	0.397
L + Al3Ni2 → Al3Ni	peritectic	853.9	0.160	0.355	0.250
NiAl + Ni3Al → Al3Ni5	peritectoid	700.2	0.600	0.726	0.625
L → (Al) + Al3Ni	eutectic	651.2	0.016	0.000	0.250

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Al–Ni).

	Structure			Stability range / °C		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	x(Ni)	at T / °C
(Al)	cF4	A1	Cu	< 27[a]	660.3	0.000	647
Al3Ni	oP16	$D0_{11}$	Fe3C	< 27[a]	853.9	0.250	77
Al3Ni2	hP5	$D5_{13}$	Ni2Al3	< 27[a]	1139.1	0.355–0.413	857

Table 2 (continued).

	Structure			Stability range / °C		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	x(Ni)	at T / °C
NiAl	cP2	B2	CsCl	< 27[a]	1649.1[b]	0.456–0.711	1367
Al3Ni5	–	–	–	< 27[a]	700.2	0.625	27
Ni3Al	cP4	$L1_2$	Cu3Au	< 27[a]	1369.6[b]	0.724–0.771	1047
(Ni)	cF4	A1	Cu	< 27[a]	1455.2[b]	0.784–1.000	1367

[a] Stable at the lowest calculated temperature.
[b] Congruent point (melting or transformation).

Al3Ni5 is stable up to 973.4 K. The solubility of Al in (Ni) reaches a mole fraction of 0.216 at 1640.0 K.

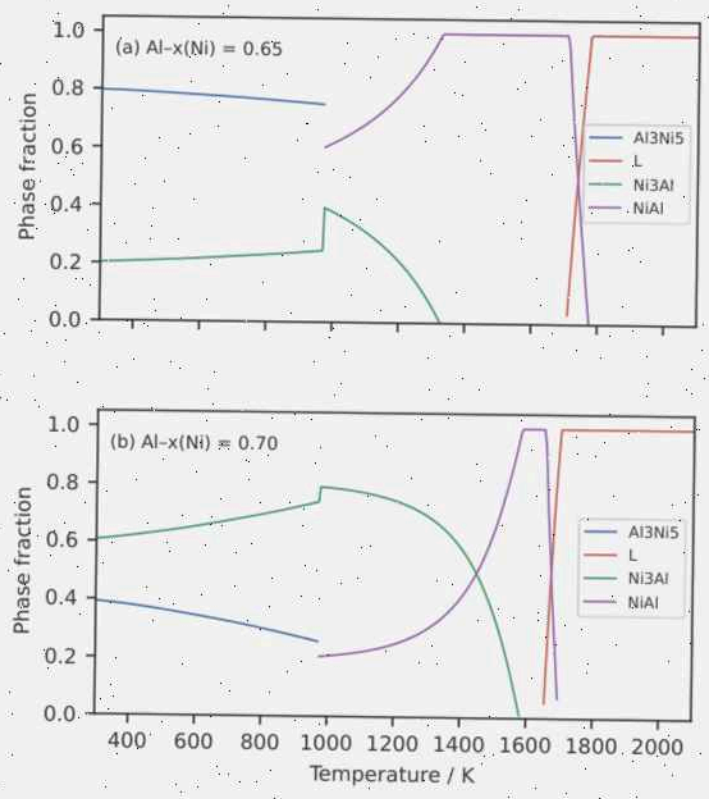


Fig. 2. Calculated equilibrium phase fractions versus temperature for Al–x(Ni) = 0.65 and x(Ni) = 0.70.

4. Thermodynamic properties

At 2000 K the integral mixing enthalpy of the liquid has its minimum, −49.27 kJ/mol, at x(Ni) = 0.500. The calculated enthalpy of formation of Al3Ni at 298.15 K is −43.45 kJ/mol per mole of atoms. Its heat capacity at room temperature is 24.46 J/(mol K). For the equiatomic liquid at 2000 K, the Gibbs energy of mixing is −45.62 kJ/mol-atom.

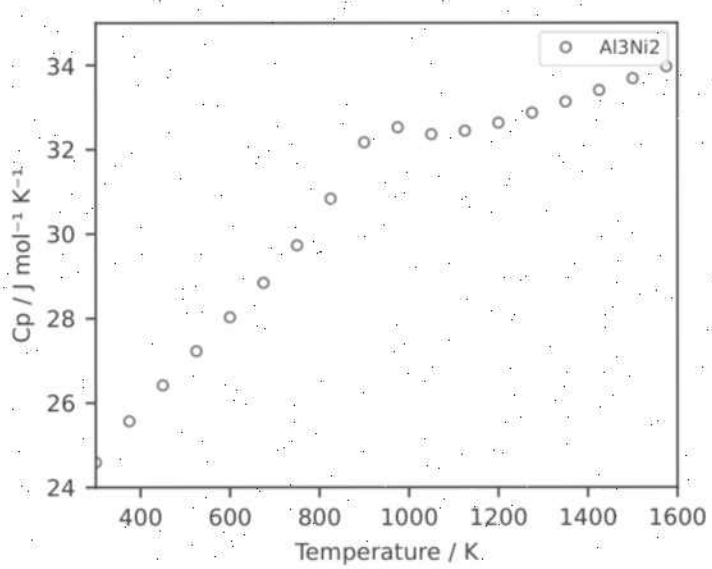


Fig. 3. Calculated heat capacity of Al3Ni2.

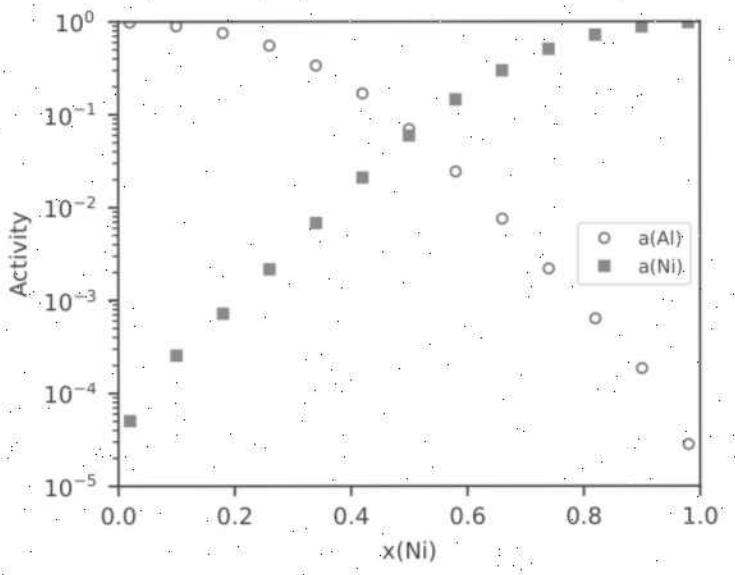


Fig. 4. Calculated activities in the liquid at 2000 K (reference: pure liquids).

Table 3. Calculated formation properties of the intermediate phases at 298.15 K (Al–Ni).

Phase	at.% Ni	ΔfH(298.15 K) / J mol⁻¹ (f.u.)	ΔfS / J mol⁻¹ K⁻¹ (f.u.)	Cp(298.15 K) / J mol⁻¹ K⁻¹ (f.u.)
Al3Ni2	39.5	-301381	-36.32	122.82
Al3Ni5	62.5	-427141	-26.19	197.79
Al3Ni	25.0	-173793	-30.14	97.86
NiAl	53.4	-124483	-8.73	49.32
Ni3Al	74.8	-160945	-6.70	99.28

Reference: stable elements (SER) at 298.15 K; values per mole of formula units (formula unit as written in the Phase column).

Table 4. Integral and partial mixing properties of the liquid at 2000 K (Al–Ni).

			Liquid at 2000 K			
x(Ni)	ΔmixH / kJ mol⁻¹	ΔmixG / kJ mol⁻¹	a(Al)	a(Ni)	ΔH(Al) / kJ mol⁻¹	ΔH(Ni) / kJ mol⁻¹
0.100	-14.14	-15.40	0.8965	2.550E-4	0.13	-142.57
0.300	-39.06	-37.24	0.4430	0.0038	-10.45	-105.83
0.500	-49.27	-45.62	0.0698	0.0593	-47.91	-50.63
0.700	-39.98	-38.15	0.0041	0.3984	-104.76	-12.21
0.900	-14.93	-16.18	1.854E-4	0.8813	-147.87	-0.15

Reference states: pure liquid Al and Ni at 2000 K; per mole of atoms. ΔH: partial molar enthalpies of mixing.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] J. Zhu, M.S. Titus, T.M. Pollock, J. Phase Equilib. Diffus., 35, 595-611(2014).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.